

2025.12.26 数学

设函数 $f(x) = \begin{cases} x^2, & 0 \leq x \leq 1, \\ 2-x, & 1 < x \leq 2. \end{cases}$ 记 $F(x) = \int_0^x f(t)dt, 0 \leq x \leq 2$, 则 $F(x) = (\quad)$.

(A) $F(x) = \begin{cases} \frac{1}{3}x^3, & 0 \leq x \leq 1, \\ \frac{1}{3} + 2x - \frac{x^2}{2}, & 1 < x \leq 2; \end{cases}$

(B) $F(x) = \begin{cases} \frac{1}{3}x^3, & 0 \leq x \leq 1, \\ \frac{7}{6} + 2x - \frac{x^2}{2}, & 1 < x \leq 2; \end{cases}$

(C) $F(x) = \begin{cases} \frac{1}{3}x^3, & 0 \leq x \leq 1, \\ \frac{x^3}{3} + 2x - \frac{x^2}{2}, & 1 < x \leq 2; \end{cases}$

(D) $F(x) = \begin{cases} \frac{1}{3}x^3, & 0 \leq x \leq 1, \\ 2x - \frac{x^2}{2}, & 1 < x \leq 2. \end{cases}$

(2) 设函数 $f(x)$ 在 $[0,1]$ 上二阶可导,且 $\int_0^1 f(x)dx = 0$, 则

(A) 当 $f'(x) < 0$ 时, $f(\frac{1}{2}) < 0$. (B) 当 $f'(x) < 0$ 时, $f(\frac{1}{2}) < 0$.

(C) 当 $f'(x) > 0$ 时, $f(\frac{1}{2}) < 0$. (D) 当 $f'(x) > 0$ 时, $f(\frac{1}{2}) < 0$.

87 设 $(-\infty, +\infty)$ 上的非负连续函数 $f(x)$ 满足 $f(x)f(1-x) = 1$, 则 $\int_0^1 \frac{x - \frac{1}{2}}{1 + f(x)} dx =$

(A) $\frac{1}{16}$ (B) $\frac{1}{8}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$

$$\begin{aligned} \int_0^1 |x - \frac{1}{2}| dx &= \int_0^{\frac{1}{2}} (\frac{1}{2} - x) dx + \int_{\frac{1}{2}}^1 (x - \frac{1}{2}) dx \\ &= 2 \int_{\frac{1}{2}}^1 (x - \frac{1}{2}) dx \\ &= 2 (\frac{1}{2}x^2 - \frac{1}{2}x) \Big|_{\frac{1}{2}}^1 \\ &= x^2 - x \Big|_{\frac{1}{2}}^1 = \frac{1}{4} \end{aligned}$$

设 $f(x)$ 为连续函数, 求 $\frac{d}{dx} \int_0^x t f(x^2 - t^2) dt$

A. $xf(x^2)$
B. $f(x^2)$
C. x^2
D. $tf(x^2 - x^2)$

$$\begin{aligned} &= \frac{d}{dx} \int_0^{x^2} f(u^2) d(u^2) \\ &= -\frac{1}{2} \frac{d}{dx} (F(x^2 + t^2)) \Big|_0^{x^2} \\ &= -\frac{1}{2} \frac{d}{dx} (F(u) - F(x)) \\ &= x f(x^2) \end{aligned}$$

设函数 $f(x)$ 与 $g(x)$ 在 $[0,1]$ 上连续, 且 $f(x) \leq g(x)$, 且对任何 $c \in (0,1)$, ()

A. $\int_{\frac{1}{2}}^c f(t)dt \geq \int_{\frac{1}{2}}^c g(t)dt$
B. $\int_{\frac{1}{2}}^c f(t)dt \leq \int_{\frac{1}{2}}^c g(t)dt$
C. $\int_c^1 f(t)dt \geq \int_c^1 g(t)dt$
D. $\int_c^1 f(t)dt \leq \int_c^1 g(t)dt$

5. 设 $x \rightarrow 0$ 时, $F(x) = \int_0^x (x^2 - t^2) f'(t) dt$ 的导数与 x^2 是等价无穷小, 试求 $f'(0)$.

$$\begin{aligned} \frac{dF(x)}{dx} &= \frac{d}{dx} \int_0^{x^2} (x^2 - t^2) f'(t) dt \\ &= \frac{d}{dx} (x^2 - t^2) f'(t) \Big|_0^{x^2} + \int_0^{x^2} 2t f'(t) dt \\ &= \frac{d}{dx} (0 - x^2 f'(0) + \int_0^{x^2} 2t f'(t) dt) \\ &= \frac{d}{dx} (-x^2 f'(0) + 2t f(t) \Big|_0^{x^2} - 2 \int_0^{x^2} t f(t) dt) \\ &= \frac{d}{dx} (-x^2 f'(0) + 2x f(x) - 2 \int_0^x t f(t) dt) \\ &= -2x f'(0) + 2f(x) + 2x f'(x) - 2f(x) \\ &= 2x (f'(x) - f'(0)) \quad f'(0) = \frac{1}{2} \end{aligned}$$

10. (单选题, 10分)

10. 若 $\int f'(x^2) dx = x^2 + C$, 则 $f(x) = (\quad)$

A. $\frac{6}{5}x^3 + C$ B. $\frac{9}{5}x^3 + C$ C. $x^3 + C$ D. $x + C$

A. A
B. B
C. C
D. D

$$\begin{aligned} f'(u) &= 3x^2 \\ f'(u) &= 3x^2 \quad f(u) = \frac{3}{5}x^{\frac{5}{2}} \end{aligned}$$

$$\int_0^1 \frac{\ln(1+x)}{(2-x)^2} dx$$

$$\begin{aligned} &= \int_0^1 \ln(x+1) d\frac{1}{2-x} = \ln 2 + \frac{1}{3} \int_0^1 \frac{d(2-x)}{2-x} - \frac{1}{3} \int_0^1 \frac{d(x+1)}{x+1} \\ &= \frac{\ln(x+1)}{2-x} \Big|_0^1 - \int_0^1 \frac{1}{2-x} \cdot \frac{1}{x+1} dx = \ln 2 + \frac{1}{3} \ln(2-x) \Big|_0^1 - \frac{1}{3} \ln(x+1) \Big|_0^1 \\ &= \ln 2 - \frac{1}{3} \int_0^1 (\frac{1}{2-x} + \frac{1}{x+1}) dx = \ln 2 - \frac{1}{3} \ln 2 = \frac{2}{3} \ln 2 \end{aligned}$$

设 $F(x) = \int_x^{x+2\pi} e^{\sin t} \sin t dt$, 则 $F(x) = (\quad)$.

A. 为正常数
B. 为负常数
C. 恒为零
D. 不为常数

$$\int_0^{2\pi} e^{\sin t} \sin t dt$$

$$\int_0^1 \frac{\ln(1+x)}{(2-x)^2} dx =$$

$$\int_0^1$$

1. 极限 $\lim_{n \rightarrow \infty} \frac{1}{n} [\sin \frac{\pi}{n} + \sin \frac{2\pi}{n} + \dots + \sin \frac{(n-1)\pi}{n}] = (\quad)$.

A. $\frac{1}{\pi}$ B. $\frac{2}{\pi}$
C. $\frac{3}{\pi}$ D. $\frac{4}{\pi}$

3. 设 $f(x)$ 有一个原函数为 $1 + \sin^2 x$, 求 $\int_0^{\frac{\pi}{2}} x f'(2x) dx$.

$$\begin{aligned} \int_0^{\frac{\pi}{2}} x df(x) &= \frac{1}{2} x f(x) \Big|_0^{\frac{\pi}{2}} - \frac{1}{4} \int_0^{\frac{\pi}{2}} f(x) dx \\ &= \frac{1}{2} x \sin x \Big|_0^{\frac{\pi}{2}} - \frac{1}{4} (1 + \sin^2 x) \Big|_0^{\frac{\pi}{2}} \\ &= 0 \end{aligned}$$

$$f(x) = 2 \sin x \cos x = \sin 2x$$

$$\int_0^1 \frac{\arcsin \sqrt{x}}{\sqrt{x(1-x)}} dx = (\quad)$$

A. $\frac{\pi^2}{4}$ B. $\frac{\pi^2}{8}$ C. $\frac{\pi}{4}$ D. $\frac{\pi}{8}$

$$2 \int_0^1 \frac{\arcsin t}{\sqrt{1-t^2}} dt$$

$$= 2 \int_0^1 \arcsin t d \arcsin t = \frac{\pi^2}{4}$$

已知 $f'(\ln x) = x$, 其中 $1 < x < +\infty, f(0) = 0$, 则 $f(x) = (\quad)$

A. e^x
B. $e^x - 1, 1 < x < +\infty$
C. $e^x - 1, 0 < x < +\infty$
D. $e^x, 1 < x < +\infty$

1. (中山大学, 2005 年) 设 $f(x)$ 在 $[0, +\infty)$ 上单调递增, 且只有有限个间断点, 则函数 $F(x) = \frac{1}{x} \int_0^x f(t) dt$ 在 $[0, +\infty)$ 上 ().

(A) 连续单调 (B) 连续但不单调
(C) 单调但不连续 (D) 既不连续又不单调

$$\int_0^1 \frac{\ln(1+x)}{1+x^2} dx =$$

A. 0
B. 1
C. $\frac{\pi}{8} \ln 2$
D. $\frac{\pi}{4} \ln 2$

$$\begin{aligned} &\int_0^1 \ln(x+1) d \arctan x \\ &= \ln(x+1) \arctan x \Big|_0^1 - \int_0^1 \arctan x \ln(x+1) dx \\ &= - \int_0^1 \frac{\arctan x}{x+1} dx \end{aligned}$$

设 $f(x)$ 为连续函数, 且 $F(x) = \frac{x^2}{x-a} \int_a^x f(t) dt$, 则 $\lim_{x \rightarrow a} F(x)$ 等于 ()

A. a
B. $a^2 f(a)$
C. 0
D. 不存在

【单选题】

$$\lim_{n \rightarrow \infty} \frac{1}{n^{\frac{3}{2}}} (\sqrt{n} + \sqrt{2n} + \dots + \sqrt{n^2}) = (\quad)$$

7. 设 $f(x) = e^{-x}$, 则 $\int \frac{f(\ln x)}{x} dx = (\quad)$

(A) $\frac{1}{x} + c$ (B) $\ln x + c$ (C) $-\frac{1}{x} + c$ (D) $-\ln x + c$

$$\int_5^{+\infty} \frac{dx}{x^2 - 4x + 3}$$

求由参数方程 $\begin{cases} x = a\cos^3\theta \\ y = a\sin^3\theta \end{cases}$ 所确定的函数的导数 $\frac{dy}{dx} = ($
)
A. $\tan\theta$
B. $-\tan\theta$
C. $\cot\theta$
D. $-\cot\theta$

6. 已知极限 $\lim_{x \rightarrow 0} \frac{x - \arctan x}{x^k} = c$, 其中 k, c 为常数, 且 $c \neq 0$, 则
(A) $k = 2, c = -\frac{1}{2}$ (B) $k = 2, c = \frac{1}{2}$
(C) $k = 3, c = -\frac{1}{3}$ (D) $k = 3, c = \frac{1}{3}$

$\lim_{x \rightarrow 0} \frac{1 - \cos x^2}{x^2 \sin^2 x} = ()$
A. 1
B. 0
C. $\frac{1}{2}$
D. 不存在

15. (单选题, 3.0 分) 设 $y = \arcsin x$, 则 $y' = ()$.
A. $\frac{1}{\sqrt{1-x^2}}$
B. $-\frac{1}{\sqrt{1-x^2}}$
C. $\frac{1}{1+x^2}$
D. $-\frac{1}{1+x^2}$

(1) 若 $\lim_{x \rightarrow 0} (e^x + ax^2 + bx)^{\frac{1}{x}} = 1$, 则
(A) $a = \frac{1}{2}, b = -1$, (B) $a = -\frac{1}{2}, b = -1$,
(C) $a = \frac{1}{2}, b = 1$, (D) $a = -\frac{1}{2}, b = 1$.

(15) 设 $\cos x - 1 = x \sin \alpha(x)$, 其中 $|\alpha(x)| < \frac{\pi}{2}$, 则
当 $x \rightarrow 0$ 时, $\alpha(x)$ 是[].
(A) 比 x 高阶的无穷小量
(B) 比 x 低阶的无穷小量
(C) 与 x 同阶但不等价的无穷小量
(D) 与 x 等价的无穷小量

5. 极限 $\lim_{x \rightarrow 0} \left[\frac{x^2}{(x-a)(x+b)} \right]^{\frac{1}{x}} = (\quad)$
(A) 1 (B) e
(C) e^{b-a} (D) e^{b-a}

1. 当 $x \rightarrow 0^+$ 时, 与 $\sqrt{x}$ 等价的无穷小量是().
A. $1 - e^x$ B. $\ln \frac{1+x}{1-\sqrt{x}}$ C. $\sqrt{1+\sqrt{x}} - 1$ D. $1 - \cos \sqrt{x}$;

$$\lim_{x \rightarrow 0^+} \frac{x^x - 1}{\ln x \cdot \ln(1-x)} =$$

$$\lim_{x \rightarrow 0} \frac{[\sin x - \sin(\sin x)] \sin x}{x^4} =$$

5. 当 $x \rightarrow 0$ 时, $\alpha(x), \beta(x)$ 是非零无穷小量, 则以下命题中,
①若 $\alpha(x) \sim \beta(x)$, 则 $\alpha^2(x) \sim \beta^2(x)$;
②若 $\alpha^2(x) \sim \beta^2(x)$, 则 $\alpha(x) \sim \beta(x)$;
③若 $\alpha(x) \sim \beta(x)$, 则 $\alpha(x) - \beta(x) = o(\alpha(x))$;
④若 $\alpha(x) - \beta(x) = o(\alpha(x))$, 则 $\alpha(x) \sim \beta(x)$.
真命题的序号为().
(A) ①③; (B) ①④; (C) ①③④; (D) ②③④

$$\lim_{x \rightarrow 0} \frac{\ln(1+x) - x}{\cos x - 1}$$

设 $y = x^x (x > 0)$, 则 $\frac{dy}{dx} = ()$
A. x^x
B. $x^x(1 + \ln x)$
C. x^{x-1}
D. $x^x \ln x$

$$\lim_{x \rightarrow 0} (1 + ax^2)^{\sin x} - 1 = 6, \text{ 则 } a = \text{——}.$$

(16) 当 $x \rightarrow 0^+$ 时, 若 $\ln^a(1+2x), (1 - \cos x)^{\frac{1}{a}}$ 均是比 x 高阶的无穷小量, 则 α 的取值范围是[].
(A) $(2, +\infty)$ (B) $(1, 2)$
(C) $\left(\frac{1}{2}, 1\right)$ (D) $\left(0, \frac{1}{2}\right)$

函数 $f(x) = x^2 e^{2x}$, 则 $f^{(n)}(0) = \text{——}.$

设函数 $f(x) = \begin{cases} a + \ln x, & x \geq 1 \\ 2ax - 1, & x < 1 \end{cases}$ 在 $(-\infty, +\infty)$ 上连续, 则 a 的值为()
A. -1
B. 0
C. 1
D. 2

设函数 $f(x) = (x-1)(x-2)(x-3)(x-4)$, 则方程 $f'(x) = 0$ 的实根的个数为()
A. 0
B. 1
C. 2
D. 3

$$\lim_{x \rightarrow 0} \frac{x \ln(1+x)}{1 - \cos x}$$

设由方程 $y + \cos(xy) = x$ 确定了隐函数 $y = y(x)$, 则 $\frac{dy}{dx}$ 在点 $(1, 0)$ 的值为
A. 0
B. 1
C. -1
D. 2

$$1: \lim_{n \rightarrow \infty} \left(\frac{-1+3n}{3n} \right)^n$$

$$2: \lim_{x \rightarrow 0} \frac{e^x - \sin x - \cos x - x^2}{\ln x \ln(x+1) \arctan \sin x}$$

$$3: \lim_{n \rightarrow \infty} n \left(\frac{1}{n^2} + \frac{1}{n^2+1} + \dots + \frac{1}{n^2+n(n+1)} \right)$$

$$4: \text{求 } f(x) = \frac{3x^3}{x^2+x-12} \text{ 的渐近线}$$

$$5: \begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases} \text{ 求 } t = \frac{\pi}{2} \text{ 处的切线方程和法线方程}$$

$$6: \text{求 } y = x^x \text{ 的导数}$$

$$7: \int \sin^3 x \, dx \quad \int \sin^4 x \, dx$$

$$8: \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\cos x}{1 + \sin x} + x \cos^{2023} x \right) dx$$

$$9: \text{求 } y = \frac{e^x + e^{-x}}{2} \text{ 从 } x=0 \text{ 到 } x=1 \text{ 的弧长}$$

$$10: \int_0^1 \frac{e^x - 1}{\sin^3 x} dx \text{ 是收敛还是发散}$$

$$1: f(x), g(x) \text{ 在 } [a, b] \text{ 连续, } f(a) > g(a), f(b) < g(b) \\ \text{证明 } \exists c \in (a, b) \text{ 使 } f(c) = g(c)$$

$$2: \text{设 } f(x), g(x) \text{ 为 } D \text{ 上的有界函数, 证明 } \sup_{x \in D} f(x) + \inf_{x \in D} g(x) \leq \sup_{x \in D} (f(x) + g(x))$$

$$3: \text{设函数 } f(x) \text{ 在 } [a, b] \text{ 连续, 在 } (a, b) \text{ 可导, } ab > 0 \\ \text{证 } \exists \xi \in (a, b) \text{ 使 } f(b) - f(a) = \frac{b-a}{ab} \xi^2 f(\xi)$$

$$4: \text{设函数 } f(x) \text{ 在 } [a, b] \text{ 上可导, 且 } f'(x) > 0 \\ \text{证明 } \int_a^b f(x) dx > (b-a) f\left(\frac{a+b}{2}\right)$$